

THESIS TOPIC PROPOSAL

CS 124 – Thesis 1

Machine Learning Classification of AI-Generated and Real Text-Containing Images Using Visual Text-Region Features

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Proposed Title

Machine Learning Classification of AI-Generated and Real Text-Containing Images Using Visual Text-Region Features

CS Domain	Machine Learning and Artificial Intelligence, with Computer Vision and Pattern Recognition
Application Area	Digital media checking and image analysis
Computing Contribution	A machine-learning approach that uses the visible features of text regions to classify an image as real or AI-generated.
Bicol University Research and Development Agenda	Social Development Promotion
Sustainable Development Goal	SDG 16 - Peace, Justice and Strong Institutions
Beneficiaries	Content reviewers, digital-media researchers, Computer Science students, organizations that handle online posts or announcements, and future researchers
Expected Innovation	The study focuses on how text looks in an image without using the meaning of the words as a basis for classification.

Background of the Study

Generative artificial intelligence can now create realistic digital images. These images may be useful for design, education, and entertainment. However, they may also be used for fake announcements, misleading posts, fabricated documents, and other false digital content. Because of this, identifying whether an image is real or AI-generated is an important problem in Computer Science.

AI-generated image detection is still difficult. An image may look realistic, or it may come from a source that was not included in the training data. Previous studies have shown that the performance of image detectors can change depending on the image source and testing condition (Zhu et al., 2023; Ojha et al., 2023; Yan et al., 2025). Therefore, results from one dataset should not automatically be considered effective for all types of images.

Images with a large amount of text may be especially difficult to classify because they contain many letters, strokes, edges, and repeated shapes. The TextFake benchmark reported that detection performance may decrease on text-rich images and referred to this problem as the "Text Density Curse" (Zhang et al., 2026).

Connection to Sustainable Development Goals, Research Agendas, and Strategic Priorities

This study supports SDG 16 by contributing to the responsible checking of digital information. It also supports the Bicol University Research and Development Agenda on Social Development Promotion because it addresses a current concern in digital communication. The proposed output is a small research prototype for preliminary screening. It will not be presented as a universal detector or as final proof of image authenticity.

Existing Challenges, Limitations, and Research Gaps

Many existing image-detection methods examine the whole image, including its texture, frequency patterns, noise, or other general visual details. These methods are useful, but they do not always show how the visible text itself affects classification.

Recent studies have identified text-rich images as a difficult setting. However, the usefulness of visual features taken specifically from text regions, without analyzing the meaning of the words, has not been sufficiently examined in a small and controlled classification study.

This research will address this gap by preparing a dataset of real and AI-generated text-containing images, extracting visual features from their text regions, and comparing selected machine-learning classifiers. The study will only make conclusions based on the selected dataset and testing conditions.

Opportunities and Proposed Computing Approach

The study will use existing image-processing and machine-learning tools. First, the visible text regions will be located and cropped. The appearance of these regions will then be measured using selected visual features. The feature values will be given to selected classifiers, such as Logistic Regression, Support Vector Machine, or Random Forest (these classifiers are just the initial choice but may be finalized in the chapter 3), to predict whether the image is real or AI-generated.

The study will mainly use publicly available datasets with real and AI-generated images. Candidate sources may include TextRich, TextFake, or other suitable public datasets. The final dataset will be selected based on accessibility, image quality, labels, usage permissions, and relevance to the study. Creating additional images may be considered only if it becomes necessary and feasible.

Discussion of Computing Components and Research Areas

The first component is text-region detection, which identifies the parts of an image that contain visible text. These parts can then be cropped for analysis.

The second component is visual feature extraction. This measures how the text appears, including its density, thickness, character shape, spacing, alignment, edge quality, texture,

contrast, and sharpness. The study will focus on the appearance of the text, not on what the text says.

The third component is machine-learning classification. Selected classifiers will use the extracted measurements to predict one of two classes: real or AI-generated. These components are related to Computer Vision, Image Processing, Pattern Recognition, and Digital Image Forensics.

Concluding Synthesis

AI-generated image detection remains a difficult Computer Science problem, especially for images that contain a large amount of text. Existing studies provide useful benchmarks and show that detector performance may change across different sources and conditions. However, the usefulness of visual features taken specifically from text regions still needs further testing. This study will examine these features using selected classifiers and a simple prototype under controlled conditions.

Introduction of the Proposed System

Core Idea and Computing Innovation

The proposed system will classify text-containing images as real or AI-generated. Its main idea is to examine the appearance of the text regions instead of relying mainly on the whole image or the meaning of the words.

The study will combine basic image processing with machine learning. It will locate text regions, measure their visual properties, and use those measurements for classification. The study will not create a new deep-learning architecture. Its main contribution is the focused evaluation of visual features from text regions.

Intended Functionality and System Components

The prototype will accept a text-containing image as input. It will locate and crop the visible text regions, extract the selected visual features, and send the results to a trained classifier.

The system will then display whether the image is predicted to be real or AI-generated. It may also display selected feature values to help users understand the process. The main components are image input, text-region processing, feature extraction, classification, and result display.

Problem Addressed and Expected Benefits

The proposed system addresses the limited understanding of how the visible text in an image may affect real-versus-AI-generated classification. It will provide a simple way to test text-region features without using semantic analysis.

Content reviewers and fact-checking practitioners may use the prototype as a preliminary screening aid. Digital-media researchers may use the results as a reference for studying text-rich

images. Computer Science students may also use the project as an example of image processing, feature extraction, and machine learning.

Relationship to Existing Studies

The study builds on previous research on AI-generated image detection. GenImage provides a large benchmark for real-versus-AI-generated image classification (Zhu et al., 2023). Ojha et al. (2023) discussed the difficulty of detecting images from different generative models, while Yan et al. (2025) showed that realistic AI-generated images remain challenging for existing detectors. TextFake further identified text-rich images as a difficult testing setting (Zhang et al., 2026).

Unlike studies that mainly examine general image features, this study will focus on visual features taken from text regions. It will provide a small and controlled evaluation rather than claim to solve universal image detection.

Concluding Overview

The proposed framework will connect text-region detection, visual feature extraction, and machine-learning classification. It will be used to evaluate whether the appearance of text regions can provide useful information for classifying real and AI-generated images. The study will also document the limitations observed in the selected dataset.

General Objective

To develop and evaluate a machine-learning approach that uses visual text-region features to classify real and AI-generated text-containing images.

Specific Objectives

1. To identify and prepare a suitable public dataset of labeled real and AI-generated text-containing images;
2. To identify and extract selected visual text-region features and implement selected machine-learning classifiers;
3. To develop a simple framework that accepts an image and provides a real or AI-generated prediction; and
4. To evaluate the classifiers using accuracy, precision, recall, F1-score, and confusion matrix.

Statement of the Problem

This study aims to determine whether visual features extracted from text regions can be used to classify real and AI-generated text-containing images.

Specifically, it seeks to answer the following questions:

1. What visual features can be extracted from the text regions of selected text-containing images?
2. How accurately can selected machine-learning classifiers distinguish real images from AI-generated images using these features?
3. Which groups of visual text-region features provide useful information for the classification task?
4. What limitations can be observed when the approach is tested on the selected dataset?

Significance of the Research

The study may help content reviewers and digital-media researchers better understand the difficulty of checking text-containing images. It may serve as a practical example for Computer Science students who are learning image processing, feature extraction, and machine learning.

Organizations that handle online posts, announcements, or other digital materials may gain a clearer understanding of the possible uses and limits of automated image classification. Future researchers may also use the feature groups, dataset preparation process, results, and limitations as a starting point for related studies.

Scope and Limitations

Dataset Scope

The study will focus on selected digital images with visible text, such as posters, announcements, and similar text-rich visual materials, depending on data availability. It will use publicly available datasets containing labeled real and AI-generated images.

The exact dataset sources, number of images, and final composition will be determined in Chapter 3. Data preparation may include checking labels, filtering text-containing images, locating text regions, cropping the regions, and preparing the feature records.

Algorithm and Model Scope

The study will use visual text-region features such as text density, stroke appearance, character shape, spacing, alignment, edge quality, texture, contrast, and sharpness.

Selected traditional machine-learning classifiers, such as Logistic Regression, Support Vector Machine, and Random Forest, will be implemented and compared. The final list will be confirmed based on feasibility. The study will not create a new deep-learning architecture and will not use the meaning, spelling, topic, or semantic content of the text.

System Scope

The system will be a simple demonstration prototype. It will allow the user to upload an image, locate and process its text regions, extract visual features, classify the image, and view the predicted class. Selected feature values may also be displayed.

The prototype is intended for research demonstration and preliminary screening. It will not be developed as a production-level public service.

Evaluation Scope

The classifiers will be evaluated using accuracy, precision, recall, F1-score, and confusion matrix. The prototype will also be checked for basic functionality, including image input, text-region processing, feature extraction, prediction, and result display.

Limitations of the Study

The results will depend on the size, quality, labels, and composition of the selected public dataset. The study is limited to binary classification: real or AI-generated. It will not identify the exact image generator, prove image provenance, or determine legal authenticity.

Unseen-generator testing, broad cross-generator testing, multilingual comparison, and image-degradation experiments such as blur, resizing, and JPEG compression are outside the initial scope. The findings will apply only to the selected dataset and controlled testing conditions.

X. Initial References

- Ojha, U., Li, Y., & Lee, Y. J. (2023). *Towards universal fake image detectors that generalize across generative models*. <https://arxiv.org/abs/2302.10174>
- Yan, S., et al. (2025). *A sanity check for AI-generated image detection*. https://proceedings.iclr.cc/paper_files/paper/2025/file/b0303773962ea1b5394c3a83cc7dd066-Paper-Conference.pdf
- Zhang, Y., et al. (2026). *TextFake: Benchmarking AI-generated image detection on text-rich images*. <https://arxiv.org/abs/2606.01050>
- Zhu, M., et al. (2023). *GenImage: A million-scale benchmark for detecting AI-generated image*. https://proceedings.neurips.cc/paper_files/paper/2023/hash/f4d4a021f9051a6c18183b059117e8b5-Abstract-Datasets_and_Benchmarks.html